

Invariant Estimator Framework for Autonomous Cars

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I. STATE ON $SE_2(3)$

We define the state on the group manifold $SE_2(3)$, such that we can maintain invariant observer properties while keeping the system as observable as possible:

$$X = \begin{bmatrix} {}^{\mathcal{W}}R_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{p}_{\mathcal{B}} \\ \mathbf{0}_{3 \times 3} & I & 0 \\ \mathbf{0}_{3 \times 3} & 0 & I \end{bmatrix} \in SE_2(3), \quad \boldsymbol{\theta} = (\mathbf{b}_g, \mathbf{b}_a) \in \mathbb{R}^6 \quad (1)$$

With the associated tangent vector defined as the following:

$$\boldsymbol{\xi} = [\boldsymbol{\xi}_{\phi} \quad \boldsymbol{\xi}_{\mathbf{v}} \quad \boldsymbol{\xi}_{\mathbf{p}} \quad \boldsymbol{\zeta}_g \quad \boldsymbol{\zeta}_a] \in \mathbb{R}^{15} \quad (2)$$